Optimal Trading Filters: a Wiener–Hopf Approach

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Keywords. Optimal trading; market impact; propagator model; inventory risk; Wiener–Hopf factorization; linear filters; fractional calculus; adapted stochastic control.
JEL classification. C61; G11; G12; G14.
MSC classification. 45E10; 60G35; 91G80; 26A33.

Abstract

A trader holding a predictive signal must weigh the expected gain against the cost of acquiring the position and the risk of holding it. Under the propagator model of market impact, trading-induced price changes involve the entire history of past trades. Existing treatments of the resulting path-wise quadratic program characterize the optimum implicitly, as the solution of an integral equation or an equivalent forward–backward system. In this paper I show how stationary solutions can be computed analytically from the signal and its forecasts using a Wiener-Hopf factorization of the friction operator. For rational frictions — inventory risk, temporary cost, and exponential resilience — the trading filter reduces to finitely many exponential moving averages, and we recover the Neuman–Voß solution away from horizon boundaries. In the (empirically more realistic) case of scale-free impact kernels, the optimal trading filter decays algebraically rather than exponentially at large frequencies; in the limit of small risk aversion, the target position is a fractional integral of the signal. Unlike its exponential (Obizhaeva-Wang) cousin, this solution never surfs its own impact or generates block trades.

1 Introduction

1.1 The quadratic optimal trading problem

A trader observes a return signal μ_t, the conditional expectation at time t of the asset's instantaneous drift, and chooses a position x_t that is bought or sold at the rate $u_t = \dot{x}_t$. Acting on the signal is costly in three ways that this paper takes as given. Holding a large position exposes capital to risk, penalized by an inventory term quadratic in x_t. Demanding immediate liquidity costs a temporary impact, quadratic in the rate u_t. And each trade shifts the price against the trades that follow it, a transient displacement that relaxes through a propagator kernel g. The trader chooses x to maximize

$$J(x) \;=\; \underbrace{\mathbb{E}\int x_t\mu_t\,dt}_{\text{gain}} \;-\; \underbrace{\frac{\lambda}{2}\,\mathbb{E}\int x_t^2\,dt}_{\text{inventory risk}} \;-\; \underbrace{\frac{\eta}{2}\,\mathbb{E}\int \dot{x}_t^2\,dt}_{\text{temporary cost}} \;-\; \underbrace{\frac{\gamma}{2}\,\mathbb{E}\iint g(|t-s|)\,\dot{x}_t\dot{x}_s\,dt\,ds}_{\text{transient cost}}, \tag{1}$$

where $\eta, \gamma, \lambda \geq 0$ weight the temporary, transient, and inventory costs. The first term is the mean–variance gain of portfolio choice; the remaining three are the frictions of execution. Sections 2–3

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treat this problem on the whole line, appropriate to an ongoing operation; Section 4 restricts it to a window $[0,T]$ closing flat, $x_T = 0$.

The difficulty in (1) comes entirely from the transient cost; the inventory and temporary terms are local in time. Following Almgren and Chriss (2001), early execution models split impact into a temporary part felt only by the trade that causes it and a permanent part linear in cumulative volume. Transaction data revised this picture: impact is concave in size (Lillo et al. 2003) and transient, a displacement that decays over time (Bouchaud et al. 2004). In the resulting propagator model the mid-price carries a weighted memory of past order flow through the kernel g, and the modeling choice reduces to the shape of g. Obizhaeva and Wang (2013) derive an exponential kernel $g(t) = e^{-\kappa t}$ from a block-shaped limit order book that refills at a constant rate, a single resilience time $1/\kappa$. Empirical decay is slower: a power law $g(t) \propto t^{-\beta}$ with β around 0.2–0.6 (Bouchaud et al. 2004, Jusselin and Rosenbaum 2020), consistent with the absence of price manipulation (Gatheral 2010) and, through the exponent, with rough volatility (Jusselin and Rosenbaum 2020). The exponential kernel is the tractable choice, the power law the empirically supported one; both are treated below.

It is convenient to refer all three costs to the position. The quadratic form in (1) is $\frac{1}{2}\langle x, Nx\rangle$ for the *friction operator*

$$N \;=\; -\eta\,\partial_t^2 \;-\; \gamma\,\partial_t^2(g * \cdot\,) \;+\; \lambda I, \qquad \hat{n}(\omega) \;=\; \eta\,\omega^2 \;+\; \gamma\,\hat{g}(\omega)\,\omega^2 \;+\; \lambda, \tag{2}$$

where $\hat{n}$ is its Fourier symbol and $\hat{g}(\omega) = \int e^{i\omega t} g(|t|)\,dt \geq 0$. Were the whole path of μ available at once, the first-order condition of (1) would be the single inversion

$$x \;=\; N^{-1}\mu, \tag{3}$$

a continuous-time analogue of the Markowitz rule $w = \lambda^{-1}\Sigma^{-1}\mu$ with the friction operator in place of the covariance matrix. The path of μ is not available at once, and this is where the difficulty lies.

1.2 The adaptedness constraint

The position at time t may depend on the signal observed up to t and not on its future: x_t must be measurable with respect to the information $\mathcal{F}_t$. When the friction is local this constraint is slack. With inventory cost alone ($\eta = \gamma = 0$) the symbol is the constant λ; reading (3) pointwise gives the memoryless Markowitz position $x_t = \mu_t/\lambda$ (Markowitz 1952, Merton 1971). Adding a temporary cost turns N into the differential operator $\lambda - \eta\,\partial_t^2$, still local in time; the optimum becomes a partial-adjustment rule that tracks a discounted target, the aim portfolio of Gârleanu and Pedersen (2013, 2016). In both cases the adapted optimum coincides with the pointwise reading of (3), and no genuine intertemporal coupling arises.

Transient impact breaks this locality. Its contribution to the symbol, $\gamma\hat{g}(\omega)\,\omega^2$, is not a polynomial, so applying N couples the position across both the past and the future. Perfect-foresight inversion (3) is then acausal, and the constraint binds. Varying (1) over adapted perturbations replaces the plain first-order condition by its projection,

$$\mathbb{E}_t\big[(Nx^\star)_t\big] \;=\; \mu_t \qquad \Longleftrightarrow \qquad P_+\,N\,P_+\,x^\star = \mu, \qquad (P_+X)_t = \mathbb{E}_t[X_t], \tag{4}$$

where P_+ is the optional projection onto adapted processes. The solution of (4) requires inverting the compressed operator P_+NP_+. Because N couples different times, its compression does not invert to the compression of N^{-1}; truncating the acausal solution to the past is not optimal.

The execution literature has resolved this coupling case by case, and largely through dynamic programming and integral equations. Deterministic liquidation under a general propagator was solved by Gatheral et al. (2012). Signal-adaptive trading was introduced for Almgren–Chriss costs by Cartea and Jaimungal (2013) and developed under temporary and exponential-resilience costs by Lehalle and Neuman (2019), Neuman and Voß (2022), Bank et al. (2017), and for general propagators through infinite-dimensional stochastic control and operator-resolvent methods (Abi Jaber and Neuman 2025, Abi Jaber et al. 2024b,a); power-law resilience with a Gaussian signal on a finite horizon was treated by Forde et al. (2022). These methods characterize the optimum implicitly, as a Riccati or backward-SDE system, or as a stochastic Fredholm equation, returning a feedback law in which the dependence on the signal is carried by a propagated state. The intuition for how a given signal shape maps to a trade is not easy to read off.

1.3 The Wiener–Hopf method

The projection in (4) is an instance of a classical obstruction. Wiener and Hopf (1931) studied convolution equations on a half-line, $\int_0^\infty k(t-s)f(s)\,ds = g(t)$ for $t \geq 0$, where the Fourier transform diagonalizes the convolution on the whole line but the support constraint couples the frequencies (Krein 1962, Noble 1958). Their device is to factor the symbol,

$$\hat{k}(\omega) \;=\; \hat{k}_-(\omega)\,\hat{k}_+(\omega),$$

with $\hat{k}_+$ analytic and zero-free in the upper half-plane and $\hat{k}_-$ in the lower. By the Paley–Wiener theorem such analyticity is causality: $\hat{k}_+$ is the transform of a kernel supported on $t \geq 0$ and $\hat{k}_-$ of one supported on $t \leq 0$. Each factor is triangular with respect to the time order, and triangular operators can be inverted forward in time. The support constraint can then be threaded between the two.

The two long-standing applications are prediction and control. In the linear prediction of a stationary series the factored object is the spectral density, and the causal factor produces the innovations representation (Wiener 1949, Whittle 1963); the same frequency-domain factorization organizes linear–quadratic dynamic optimization (Whiteman 1985). Existence of a factorization for a general positive operator, beyond the scalar convolution case, was established by Gohberg and Krein (1970) for Volterra operators on an interval and by Arveson (1975) for positive operators relative to a nest of subspaces. These theorems assert the factorization; they do not, on their own, compute the factor. In the stationary case the factor is explicit, given by the Szegő outer function of the symbol.

The present paper applies this device to the trading problem, with the filtration playing the role of the half-line. Causality of a trading strategy is adaptedness, and the nest of subspaces is the increasing family of adapted spaces indexed by $\mathcal{F}_t$. The object to factor is the friction N itself. Splitting $N = N_- N_+$ into a causal factor and its anticausal adjoint gives a formula for the compressed inverse $(P_+ N P_+)^{-1}$, and the optimal position follows in closed form from the signal and its forecasts.

1.4 Contribution and outline

Section 2 gives the whole-line solution. A projected-inverse identity (Lemma 1) expresses the compressed inverse of any factored positive operator through its one-sided factors, and yields the adapted optimum for a general adapted signal in terms of its forecast curve (Theorem 1). For a stationary Gaussian signal the optimum becomes an explicit linear filter of the signal (Theorem 2), obtained by a two-fold factorization — of the friction and of the signal. The construction also returns the value the trader forgoes to the constraint, as the energy of a nonanticipativity multiplier, and a benchmark against the perfect-foresight optimum.

Section 3 works out the filter for the two kernel families. Rational frictions — inventory risk, temporary cost, exponential resilience, and their sums — give a filter that is a finite sum of exponential moving averages of the signal, recovering the Markowitz rule, the aim portfolio, and the resilience models of Lehalle and Neuman (2019) and Neuman and Voß (2022) as low-order cases. The scale-free power law has no rational reduction: the optimal rate is a fractional derivative of the forecast, the position a fractional integral of the return, and the value kept under adaptedness is a fixed fraction $\sin(\pi\beta/2)$ of the perfect-foresight value.

Section 4 carries the factorization to a finite horizon. There the Szegő factor is replaced by a Gohberg–Krein Volterra factor anchored at the terminal time, the interior relaxes to the stationary filter with an explicitly bounded boundary layer (Proposition 1), and the pure-impact problems with a signal admit closed forms: block-plus-continuous liquidation for the exponential kernel, and a continuous, U-shaped profile for the power law. Section 5 states the scope and its limits. Proofs are collected in the appendix.

2 The Wiener–Hopf solution

Fix a filtered probability space satisfying the usual conditions. The signal μ is adapted and mean-zero, with forecast curve

$$\bar\mu(t,s) \;=\; \mathbb{E}_t[\mu_s] \quad (s \ge t), \qquad \bar\mu(t,s) = \mu_s \quad (s \le t),$$

decaying to zero as $s \to \infty$. The Fourier convention is $\hat f(\omega) = \int e^{i\omega t} f(t)\,dt$, under which causal kernels have transforms analytic in the upper half-plane. Admissible positions are adapted with finite friction energy $\mathbb{E}\langle x, Nx\rangle < \infty$.

2.1 Factorization of the friction

The symbol $\hat n$ in (2) is positive, even, and of polynomial growth, and it satisfies the Szegő condition $\int |\log \hat n(\omega)|\,(1+\omega^2)^{-1}\,d\omega < \infty$. This holds for every kernel considered here, including the pure power law: there $\hat n$ vanishes at the origin as $|\omega|^{1+\beta}$, a zero that enters $\log \hat n$ only logarithmically and remains integrable (Krein 1962). Log-integrability is what permits the factorization.

Assumption 1 (Friction)**.** The weights satisfy $\eta, \gamma, \lambda \ge 0$; the propagator g is locally integrable and positive-definite, so $\hat g \ge 0$; and $\hat n$ is positive almost everywhere and satisfies the Szegő condition.

Under Assumption 1 the friction factors as $N = N_- N_+$, with symbol

$$\hat n(\omega) \;=\; \hat n_-(\omega)\,\hat n_+(\omega), \qquad \hat n_- = \overline{\hat n_+}, \tag{5}$$

where $\hat n_+$ is analytic and zero-free in the upper half-plane. The factor N_+ is then causal with causal inverse, and $N_- = N_+^*$ is its anticausal adjoint. The outer factor is given by the Szegő formula

$$\hat n_+(\omega) \;=\; \exp\!\Big(\frac{1}{2\pi i}\int\Big(\frac{1}{t-\omega} - \frac{t}{1+t^2}\Big)\log \hat n(t)\,dt\Big), \qquad \operatorname{Im}\omega > 0, \tag{6}$$

unique up to a unimodular constant (Krein 1962). Evaluated on the imaginary axis, and using that $\hat n$ is even, (6) reduces to the real Poisson integral

$$\Phi(\theta) \;:=\; \hat n_+(i\theta) \;=\; \exp\Big[\frac{\theta}{2\pi}\int \frac{\log \hat n(t)}{\theta^2 + t^2}\,dt\Big] \;>\; 0, \tag{7}$$

which will carry the dependence of the filter on the signal's mean-reversion rate. Conjugating the friction inner product by N_-^{-1} flattens it to the ordinary L^2 product; this is the sense in which the factorization whitens N. When the working variable is the rate rather than the position, the same construction applies to the rate-referred friction $Q = N(-\partial_t^2)^{-1}$, with symbol $\hat{q} = \hat{n}/\omega^2$ and factors $\hat{q}_\pm = \hat{n}_\pm/(\mp i\omega)$; this is the form used for the scale-free kernel in Section 3.2 and on the finite horizon in Section 4.

2.2 The projected inverse

We can now assemble the compressed inverse required by (4) from the one-sided factors.

Lemma 1 (Adapted projected inverse)**.** *Let $A = A_- A_+$ be a positive operator with A_+ causal with causal inverse and $A_- = A_+^*$ anticausal with anticausal inverse. Then, on adapted processes,*

$$(P_+ \, A \, P_+)^{-1} \; = \; A_+^{-1} \, P_+ \, A_-^{-1}. \tag{8}$$

The proof (Appendix A) is triangular bookkeeping: A_+ and its inverse preserve the adapted subspace, A_- and its inverse preserve the orthogonal complement, and P_+ annihilates whatever falls into that complement. Lemma 1 is the projected-inverse identity behind the triangular factorization of a positive operator along a nest (Gohberg and Krein 1970, Arveson 1975); here the nest is the family of adapted subspaces, and (5) is the corresponding Cholesky factorization along the filtration.

Theorem 1 (Optimal policy for a general adapted signal)**.** *Under Assumption 1 the maximizer of* (1) *exists, is unique, and equals*

$$x^\star \; = \; N_+^{-1} \, P_+ \, N_-^{-1} \, \mu, \qquad \textit{equivalently} \qquad x_t^\star = \big(N_+^{-1}\zeta\big)_t, \quad \zeta_s = \big(N_-^{-1} \, \bar{\mu}(s,\cdot)\big)(s), \tag{9}$$

the second form using that the deterministic operator N_-^{-1} commutes with conditioning on its time argument.

Three steps make up the policy (9): first, whiten by the anticausal factor N_-^{-1}; since that would draw on the true future of μ, replace it by the forecast curve $\bar{\mu}(s,\cdot)$, which yields the whitened forecast ζ; then recolour by the causal factor N_+^{-1}. Completing the square in (1) gives an equivalent reading: up to a constant, $J(x) = -\frac{1}{2}\|x - N^{-1}\mu\|_N^2$ in the friction norm $\|y\|_N^2 = \mathbb{E}\langle y, Ny\rangle$, so $x^\star$ is the orthogonal projection of the perfect-foresight optimum $N^{-1}\mu$ onto the adapted positions in that norm. This is a Wiener–Kolmogorov estimation problem with the friction in place of the signal covariance, and the correspondence is the one that carries frequency-domain factorization from filtering into control (Wiener 1949, Whittle 1963, Whiteman 1985).

The gap between $x^\star$ and the perfect-foresight optimum has a dual description. Applying N to (9),

$$Nx^\star \; = \; \mu - \xi^\star, \qquad \xi^\star = N_- \, (I - P_+) \, N_-^{-1} \, \mu, \qquad P_+ \xi^\star = 0, \tag{10}$$

so $x^\star$ solves the *unconstrained* first-order condition for the modified signal $\mu - \xi^\star$. A correction with vanishing optional projection is the nonanticipativity multiplier of Rockafellar and Wets (1976), and (10) is its closed form. Because $\mathbb{E}\langle x, \xi^\star\rangle = 0$ for every adapted x, the value surrendered to adaptedness equals $\frac{1}{2}\mathbb{E}\langle \xi^\star, N^{-1}\xi^\star\rangle$, the friction energy of the multiplier.

2.3 The stationary filter

For a stationary Gaussian signal the forecast is itself a filter, and (9) becomes a transfer function. Let μ be purely non-deterministic and generate its own filtration, with Wold representation $\mu = \psi * \dot{W}$ and spectral density $S_\mu = |\hat{\psi}|^2$, where $\dot{W}$ is the unit white noise generating $\mathcal{F}_t$ and $\hat{\psi}$ is the outer spectral factor (Wiener 1949, Whittle 1963).

Assumption 2 (Signal)**.** The signal μ is stationary, Gaussian, purely non-deterministic, generates its own filtration, and has the Wold representation $\mu = \psi * \dot{W}$ with outer spectral factor $\hat{\psi}$.

In the innovations $\dot{W}$ the whitened signal $N_-^{-1}\mu$ has symbol $h = \hat{\psi}\,\hat{n}_-^{-1}$. Conditioning at time s discards the innovations after s, so the projection P_+ truncates the corresponding moving-average kernel at lag zero and acts on symbols as the Riesz plus-part $[\cdot]_+$ onto non-negative lags.

Theorem 2 (Optimal trading filter)**.** *Under Assumptions 1 and 2 and $\lambda > 0$, the optimal position is the filter $x^\star = \pi * \dot{W}$ with transfer function*

$$\hat{\pi} \;=\; \hat{n}_+^{-1}\,[h]_+\,, \qquad h = \hat{\psi}\,\hat{n}_-^{-1}, \tag{11}$$

the optimal rate is $\hat{\chi} = (-i\omega)\,\hat{\pi}$, and the expected profit per unit time is

$$v \;=\; \frac{1}{4\pi}\,\big\|[h]_+\big\|_{L^2}^2. \tag{12}$$

Requiring $\lambda > 0$ keeps h and $\hat{\pi}$ in L^2. Without inventory cost the symbol vanishes at the origin and the position ceases to be stationary; Section 3.2 treats that regime, referring the theory to the rate.

When the signal has a finite-dimensional state the filter simplifies further. Call μ Markov of order d if its forecast curve is a fixed combination of the current value and its first $d-1$ derivatives, $\bar{\mu}(t,s) = \sum_{k=0}^{d-1} \mu_t^{(k)}\, g_k(s-t)$; equivalently S_μ is rational of degree d. Whitening the forecast then produces a differential operator, $\zeta_s = \sum_k \rho_k \mu_s^{(k)}$ with $\rho_k = (N_-^{-1} g_k)(0)$, and on the observed signal the policy becomes $\hat{x}^\star = P(-i\omega)\,\hat{\mu}/\hat{n}_+$. The order-one case is the Ornstein–Uhlenbeck signal, the only stationary Gaussian signal with a fixed forecast profile: $\bar{\mu}(t,s) = \mu_t e^{-\theta(s-t)}$ with mean-reversion rate $\theta > 0$. With a single pole, $\rho_0 = 1/\Phi(\theta)$, and

$$\hat{x}^\star(\omega) \;=\; \frac{\hat{\mu}(\omega)}{\Phi(\theta)\,\hat{n}_+(\omega)}, \qquad v = \frac{\sigma^2\theta}{4\,\Phi(\theta)^2}, \tag{13}$$

with $\theta^2\sigma^2$ the innovation variance of μ. Every rational friction is a special case of (13), worked out in Section 3.

2.4 The value of anticipation

A benchmark for (12) comes from perfect foresight: a trader with the whole path of μ in hand replaces $[\cdot]_+$ by the identity and attains

$$v_{\text{ant}} \;=\; \frac{1}{4\pi}\int \frac{S_\mu(\omega)}{\hat{n}(\omega)}\,d\omega \;\geq\; v, \tag{14}$$

finite for $\lambda > 0$. The *causality gap* $v/v_{\text{ant}} \leq 1$ is the fraction of the perfect-foresight value that the adapted trader retains. By orthogonality of causal and anticausal parts, the shortfall equals the energy of the anticausal part of h,

$$v_{\text{ant}} - v \;=\; \frac{1}{4\pi}\big\|[h]_-\big\|_{L^2}^2 \;=\; \tfrac{1}{2}\,\mathbb{E}\langle \xi^\star, N^{-1}\xi^\star\rangle,$$

matching the multiplier energy of (10). The gap thus has a closed form. For power-law impact it takes a value that does not depend on the signal, worked out in Section 3.2.

3 Exact filters

The whole-line filter (13) is explicit once $\hat{n}_+$ is known. This section computes it for the two kernel families. Rational frictions give a finite sum of moving averages (Section 3.1); the scale-free power law gives a fractional operator (Section 3.2).

3.1 Exponential kernels: from zero to two moving averages

When the friction has a rational symbol, the outer factor $\hat{n}_+$ is a ratio of polynomials, and the position is a finite sum of exponential moving averages of the signal whose rates are the zeros of $\hat{n}_+$. The classical rules are the low-order cases.

Inventory risk alone. With $\eta = \gamma = 0$ the symbol is the constant λ, the factorization is trivial, and the position is the memoryless Markowitz rule $x_t^\star = \mu_t/\lambda$ (Markowitz 1952, Merton 1971), with value $v = \sigma^2\theta/4\lambda$ for an Ornstein–Uhlenbeck signal. Zero moving averages.

Temporary cost and risk. With $\gamma = 0$ the friction is the differential operator $\lambda - \eta\,\partial_t^2$, and $\hat{n}_+ = \sqrt{\eta}\,(a - i\omega)$ with $a = \sqrt{\lambda/\eta}$: one moving average. Differentiating (13) puts it in partial-adjustment form,

$$u_t^\star = a\,(\mathrm{aim}_t - x_t^\star), \qquad \mathrm{aim}_t = \frac{a}{a+\theta}\cdot\frac{\theta\alpha_t}{\lambda}, \tag{15}$$

trading toward a target that discounts the Markowitz position by the persistence factor $a/(a+\theta)$, with $\alpha_t = \mu_t/\theta$. This is the aim portfolio of Gârleanu and Pedersen (2016), with general targets in Bank et al. (2017).

Exponential resilience and risk. For $g = e^{-\kappa|t|}$ and $\eta = 0$, the factor is $\hat{n}_+ = \sqrt{A}\,(m - i\omega)/(\kappa - i\omega)$ with $A = 2\kappa\gamma + \lambda$ and $m = \kappa\sqrt{\lambda/A}$, and the position is the current signal plus one moving average,

$$x_t^\star \;=\; \frac{\kappa+\theta}{A(m+\theta)}\Big[\mu_t + (\kappa - m)\int_{-\infty}^{t} e^{-m(t-s)}\mu_s\,ds\Big], \tag{16}$$

whose rate m increases from 0 to κ with risk aversion. This is the stationary, no-temporary-cost form of the model of Lehalle and Neuman (2019), with risk aversion in the role their finite horizon plays.

All three frictions. For $g = e^{-\kappa|t|}$ with $\eta, \gamma, \lambda > 0$ the symbol remains rational, and $\hat{n}_+ = \sqrt{\eta}\,(b_1 - i\omega)(b_2 - i\omega)/(\kappa - i\omega)$ has two zeros, fixed by $b_1^2 b_2^2 = \lambda\kappa^2/\eta$ and $b_1^2 + b_2^2 = \kappa^2 + (2\kappa\gamma + \lambda)/\eta$. The position is a two-average filter,

$$x_t^\star = \frac{1}{\Phi(\theta)\sqrt{\eta}}\sum_{i=1}^{2} w_i \int_{-\infty}^{t} e^{-b_i(t-s)}\mu_s\,ds, \qquad w_i = \frac{\kappa - b_i}{b_j - b_i}.$$

Neuman and Voß (2022) solve this configuration through a coupled forward–backward SDE system and report a feedback control; in the stationary regime its gains reduce to these two rates and weights, and the finite-horizon gains converge to them in the interior. The two degenerations are exact: at $\gamma = 0$ the rates are $\{a, \kappa\}$ and the κ-pole cancels, recovering (15); as $\eta \to 0$ one rate tends to m and the filter recovers (16). A rational friction with n poles gives at most n moving averages. Figure 1 shows how the rates and the resulting policy depend on the friction.

3.2 Power-law kernels and fractional calculus

The power-law kernel has no characteristic timescale: its transform $\hat{g}(\omega) = c_\beta|\omega|^{\beta-1}$, with $c_\beta = 2\Gamma(1-\beta)\sin(\pi\beta/2)$, diverges at the origin, so the impact is non-integrable. Writing $|t|^{-\beta} =$

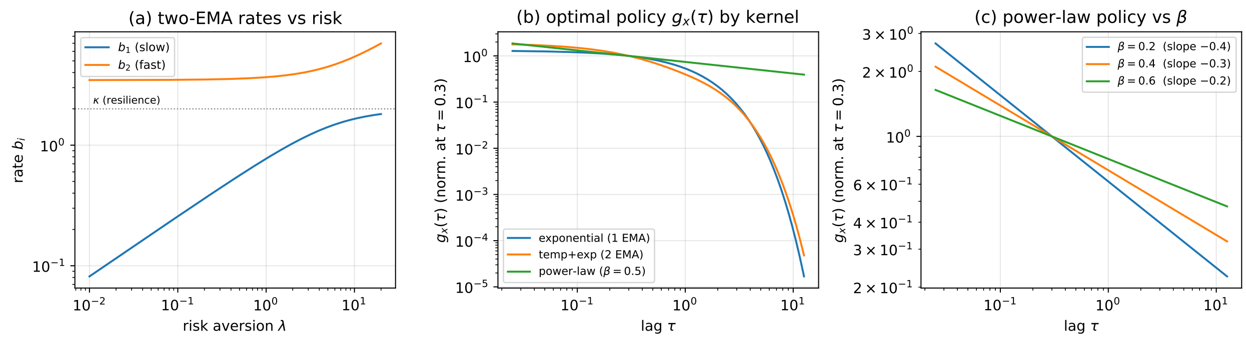

Figure 1: The optimal policy across kernels. (a) For the temporary-plus-resilient kernel ($\eta = 0.5$, $\gamma = 1$, $\kappa = 2$), the two moving-average rates b_1, b_2 against risk aversion λ: the slow rate tends to zero as $\lambda \to 0$, where the position becomes non-stationary, and both grow with λ. (b) Position impulse responses by kernel ($\theta = 1$, normalized at $\tau = 0.3$): the exponential and two-average kernels cut off exponentially, the power-law kernel keeps an algebraic tail (Section 3.2), and the Markowitz limit is a spike at zero lag. (c) The power-law policy across β, a scale-free kernel of slope $(\beta - 1)/2$.

$\Gamma(\beta)^{-1} \int_0^\infty e^{-r|t|} r^{\beta-1}\, dr$ exhibits it as a superposition of exponential kernels over a continuum of decay rates, where a rational kernel has finitely many. The optimal filter is correspondingly a fractional operator.

With $\eta = \lambda = 0$ the symbol $\hat{n} = \gamma c_\beta |\omega|^{1+\beta}$ vanishes at the origin, so the optimal position is non-stationary and the stationary object is the rate. The natural signal is then the appreciation $\alpha_t = \mathbb{E}_t \int_t^\infty \mu_s\, ds$, with $\mu = \mathbb{E}_t[-\dot{\alpha}]$; for an Ornstein–Uhlenbeck signal $\alpha = \mu/\theta$. Referred to the rate the friction is Q with symbol $\hat{q} = \gamma \hat{g}$ and factors $\hat{q}_\pm$, Lemma 1 applies to $Q = Q_- Q_+$, and Theorem 1 reads

$$u^\star \;=\; Q_+^{-1} P_+ Q_-^{-1} \alpha, \qquad u_t^\star = \big(Q_+^{-1} \zeta\big)_t, \quad \zeta_s = \big(Q_-^{-1} \bar{\alpha}(s, \cdot)\big)(s), \tag{17}$$

with the whitened forecasts agreeing across the two variables (Appendix A).

The policy is a fractional derivative. The Liouville fractional integral of order $\nu \in (0, 1)$ is

$$(I_+^\nu f)(t) = \frac{1}{\Gamma(\nu)} \int_{-\infty}^t (t - s)^{\nu - 1} f(s)\, ds,$$

with I_-^ν its time reflection; it weights the past by the kernel $(t-s)^{\nu-1}$ and interpolates between the identity at $\nu \to 0$ and a single integration at $\nu = 1$. Its inverse is the fractional derivative $D_\pm^\nu$, which for the bounded signals here takes the Marchaud form (Samko et al. 1993)

$$(D_\pm^\nu f)(t) = \frac{\nu}{\Gamma(1-\nu)} \int_0^\infty \frac{f(t) - f(t \mp s)}{s^{1+\nu}}\, ds,$$

whose increment keeps the integral finite for non-decaying f. Because $\hat{q}_\pm = (\gamma c_\beta)^{1/2} (\mp i\omega)^{-\nu}$ with $\nu = (1-\beta)/2$, the rate-referred factors are fractional integrals, $Q_\pm = (\gamma c_\beta)^{1/2} I_\pm^\nu$, and (17) becomes

$$u_t^\star \;=\; \frac{1}{\gamma c_\beta} (D_+^\nu \zeta)(t), \qquad \zeta_s = \big(D_-^\nu\, \bar{\alpha}(s, \cdot)\big)(s) : \tag{18}$$

an anticausal fractional derivative of the forecast, whitened, followed by a causal fractional derivative, both of order $\nu \in (0.2, 0.4)$ over the empirical range of β. For an Ornstein–Uhlenbeck signal $\zeta_s = \theta^\nu \alpha_s$, so $u^\star = (\theta^\nu / \gamma c_\beta)\, D_+^\nu \alpha$ with value $v = \sigma^2 \theta^{-\beta} / 4\gamma c_\beta$.

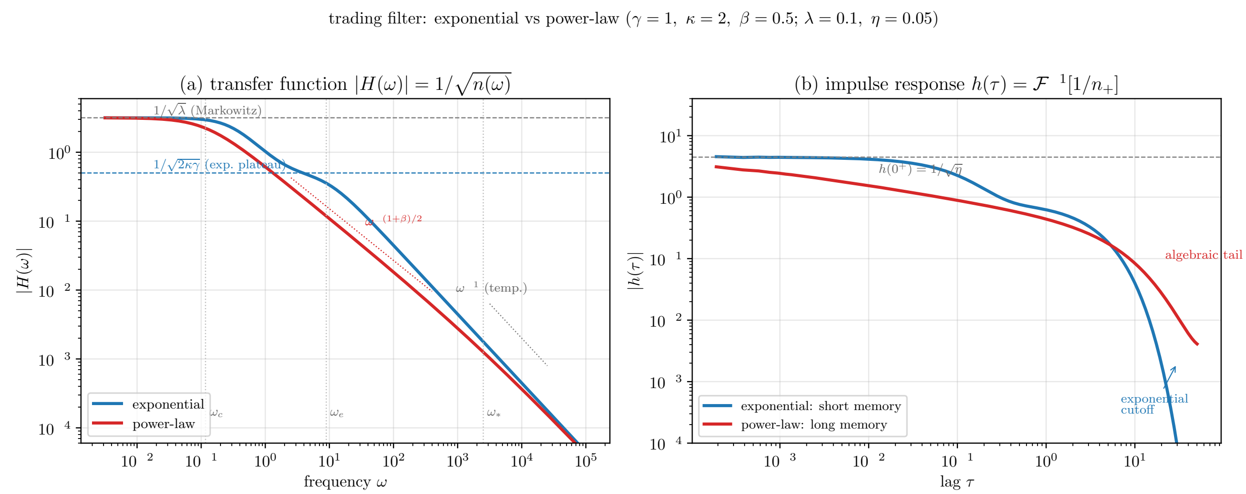

Figure 2: The trading filter for exponential versus power-law transient impact ($\gamma = 1$, $\kappa = 2$, $\beta = 0.5$; small $\lambda = 0.1$, $\eta = 0.05$). (a) Transfer function $|H(\omega)| = 1/\sqrt{\hat{n}(\omega)}$: both share the low-frequency Markowitz plateau and roll off as ω^{-1} under the temporary cost at high frequency; between them the exponential settles on a finite-memory plateau while the power law slides as $\omega^{-(1+\beta)/2}$. (b) Impulse response $h(\tau) = \mathcal{F}^{-1}[\hat{n}_+^{-1}]$: the exponential filter cuts off past the temporary-cost scale, the power law leaves an algebraic tail.

The position is a fractional integral. Accumulating this rate, the position filter (13) has $\hat{n}_+ \propto (-i\omega)^{(1+\beta)/2}$, so

$$\hat{x}^\star(\omega) \;\propto\; (-i\omega)^{-(1+\beta)/2}\,\hat{\mu}(\omega), \qquad x^\star \propto I_+^{(1+\beta)/2}\mu, \tag{19}$$

the scale-free counterpart of the Markowitz position μ/λ. Its spectral density is $\propto |\omega|^{-(1+\beta)} S_\mu(\omega)$, which is integrable at the origin only if $S_\mu(\omega) = o(|\omega|^\beta)$ there. A persistent return fails this test; for an Ornstein–Uhlenbeck signal $S_\mu(0) \neq 0$ and the position drifts without bound, a fractional analogue of the random walk that integrating a persistent return would produce. This is why the pure power-law theory is referred to the rate, which is stationary for every signal. Any inventory cost $\lambda > 0$ confines the position, at the cost of leaving the scale-free regime; the transfer function then interpolates across the crossover frequencies of $\hat{q}$, risk-dominated at low frequency and temporary-cost-dominated at high frequency, as Figure 2 shows.

The value of information

Under pure power-law impact the causality gap of Section 2.4 does not depend on the signal. The adapted value is $v = \sigma^2\theta^{-\beta}/4\gamma c_\beta$; the perfect-foresight benchmark (14), referred to the rate at $\lambda = 0$, equals $v_{\text{ant}} = v/\sin(\pi\beta/2)$, so

$$\frac{v}{v_{\text{ant}}} \;=\; \sin\frac{\pi\beta}{2}, \tag{20}$$

the sine arising from the argument of $(-i\omega)^{\beta-1}$ on the two half-lines. Adaptedness costs little for short-memory impact ($\beta \to 1$) and nearly everything for near-permanent impact ($\beta \to 0$); over the empirical range the adapted trader retains between a third and four-fifths of the perfect-foresight value (Figure 3).

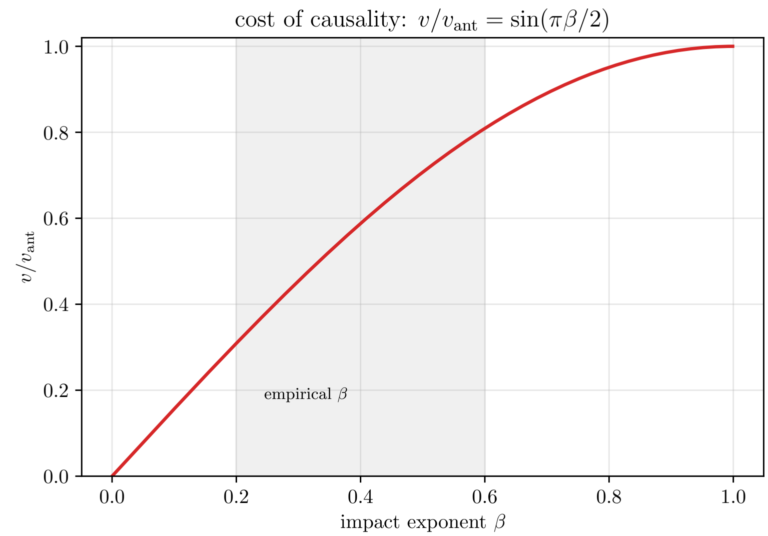

Figure 3: The causality gap under power-law impact. The adapted value is a fixed fraction $v/v_{\text{ant}} = \sin(\pi\beta/2)$ of the perfect-foresight value, independent of the signal speed; the shaded band marks the empirical range of β.

4 Finite-horizon solutions

Most execution problems are posed on a fixed window rather than the whole line: a parent order is worked over a session, or an inherited position is unwound by a deadline, ending flat. On $[0, T]$ translation invariance is lost, the signal and friction no longer share a Fourier basis, and the Szegő formula (6) no longer applies. The general finite-horizon problem has been characterized implicitly in the literature cited in Section 1.2; closed forms have been available only in the absence of a signal, for the exponential kernel (Obizhaeva and Wang 2013) and the power law (Gatheral et al. 2012). Below the factorization survives the loss of translation invariance, and the pure-impact problems admit closed forms with a signal.

4.1 The Gohberg–Krein factorization

On the window the working variable is the rate, which carries both the terminal constraint and the endpoint effects. The rate-referred friction on $[0, T]$, $G_T = \eta I$ plus a continuous-kernel integral operator, is positive. Relative to the nest of adapted subspaces such an operator factors triangularly, $G_T = C_- C_+$ with C_+ a causal Volterra operator and $C_- = C_+^*$ (Gohberg and Krein 1970, Arveson 1975). This is the finite-horizon analogue of the Szegő factorization, and the two existence theorems apply to G_T directly. Lemma 1 then holds verbatim, and the optimum is

$$u^\star \;=\; C_+^{-1}\, P_+\, C_-^{-1}\, \alpha^{\text{eff}}, \qquad \alpha^{\text{eff}} = \alpha + \sum_k \xi_k e_k, \tag{21}$$

where α^{eff} adjoins one multiplier ξ_k per linear position constraint $\langle e_k, x\rangle = 0$ — a process-valued multiplier for the pathwise constraint $x_T = 0$ — each modifying the signal within its constraint's annihilator, exactly as the nonanticipativity multiplier (10) does for adaptedness (Abi Jaber et al. 2024a). The Szegő factor is replaced by a Volterra kernel; nothing else in the construction changes. The adapted optimum is unique (Theorem 1), so (21) and the general-propagator characterizations

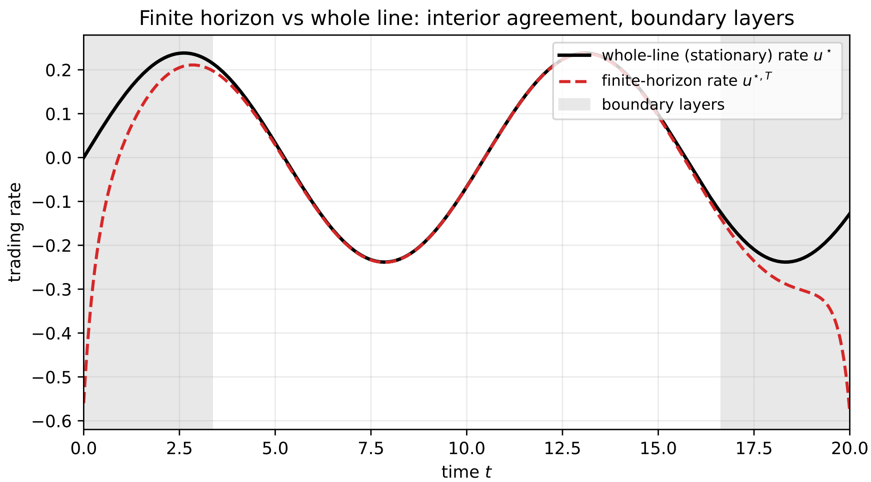

Figure 4: Finite horizon versus whole line ($\eta = 0.5$, $\gamma = 1$, $\kappa = 2$, $\lambda = 1$, $T = 20$). The finite-horizon optimal rate follows the whole-line stationary rate through the interior and departs only in the start-up and terminal boundary layers of Proposition 1; the shading marks the boundary-layer scale, not a fitted width.

of Abi Jaber and Neuman (2025), Abi Jaber et al. (2024b,a) describe one process; the iterative solution of their stochastic Fredholm equation is the Neumann expansion that the Gohberg–Krein factors sum in closed form.

The two endpoints and the terminal constraint act locally, and their influence decays into the interior, where (21) relaxes to the stationary filter of Section 2.3.

Proposition 1 (Boundary-layer decay)**.** *Let $u^{\star,T}$ be the finite-horizon optimum on $[0,T]$ and $u^\star$ the whole-line optimum for the same bounded signal, and set $d(t) = \min(t, T-t)$. Then, for constants depending only on the kernel parameters,*

$$\left|u_t^{\star,T} - u_t^\star\right| \le \begin{cases} C(\beta)\,\|\alpha\|_\infty\, d(t)^{-\nu}, & \textit{power-law kernel},\ \nu = \frac{1-\beta}{2}, \\ C\,\|\alpha\|_\infty\, e^{-b_1 d(t)}, & \textit{rational symbol},\ b_1 \textit{ the slowest zero of } \hat{n}_+. \end{cases} \tag{22}$$

The proof is in Appendix A. The boundary layer has width $1/b_1$ for a rational kernel and is set by the impact memory for a power law, thin except at the small ν of the empirical exponents. Figure 4 shows the finite-horizon rate tracking the stationary filter through the interior.

4.2 Pure transient impact with a signal

The explicit Volterra factor depends on the kernel and closes in the two cases of interest.

Exponential kernel. For $g = e^{-\kappa|t|}$ with $\eta = \lambda = 0$, the operator G_T has kernel $\gamma e^{-\kappa|t-s|}$, and its inverse factors are the local differential operators

$$C_+^{-1} = \frac{\partial_t + \kappa}{\sqrt{2\gamma\kappa}}, \qquad C_-^{-1} = \frac{\kappa - \partial_t}{\sqrt{2\gamma\kappa}}, \tag{23}$$

with the two Robin conditions $h'(0) = \kappa h(0)$ and $h'(T) = -\kappa h(T)$ carried by rank-one endpoint atoms; the causal factor itself has Green's function $\sqrt{2\gamma\kappa}\, e^{-\kappa(t-s)}$. Because the kernel is finite at zero lag, $g(0) < \infty$, an instantaneous trade costs only finitely, and the optimum pays for it with a discrete block at each endpoint, trading continuously between: the block-plus-continuous liquidation of Obizhaeva and Wang (2013), here with a signal through the projection in (21). A rational symbol of degree d factors analogously, with d atoms per endpoint.

Power-law kernel. For $g = |t|^{-\beta}$ with $\eta = \lambda = 0$, the causal factor is the terminal-anchored Volterra operator with kernel

$$c_+(t,s) = (\gamma c_\beta)^{1/2}\Big(\frac{T-s}{T-t}\Big)^{\nu}\frac{(t-s)^{\nu-1}}{\Gamma(\nu)}, \qquad 0 \le s \le t \le T, \tag{24}$$

verified by direct kernel integration to give $C_-C_+ = G_T$ with the correct constant. This is the whole-line factor $(\gamma c_\beta)^{1/2} I_+^{\nu}$ multiplied by the terminal weight $(T-t)^{\nu}$; homogeneity of the kernel is what reduces the anchoring to a scalar weight, and away from the endpoints the weight tends to one, which is Proposition 1 for this kernel. Because $g(0) = \infty$, an instantaneous trade is prohibitively costly and no endpoint blocks appear. For a constant signal the projection is trivial and (21) reduces to the continuous U-shaped profile $u^\star(t) \propto [t(T-t)]^{(\beta-1)/2}$ of Gatheral et al. (2012, Ex. 2.30). For a Gaussian Volterra signal the finite-horizon first-order condition is the stochastic Fredholm equation solved by Forde et al. (2022); (21) with (24) is its solution operator, the time reflection of (24) supplying their left-anchored factor and the generalized Abel inversion of Chakrabarti and George (1994) implementing G_T^{-1}.

5 Concluding remarks

Adaptedness binds in the trading problem because transient impact couples different times, and the compression of the friction to the past does not invert to the compression of its inverse. Factoring the friction along the filtration, into a causal part and its anticausal adjoint, resolves the coupling: the adapted inverse assembles from the one-sided factors, and the optimal policy follows from the signal and its forecasts. On the whole line the factor is a frequency-domain formula and the policy is a linear filter, whose special cases are the Markowitz rule, the aim portfolio, the moving-average filters of the resilience models, and, for scale-free impact, a fractional derivative. The shape of the policy follows the shape of the impact memory. On a finite horizon the factor is a terminal-anchored Volterra kernel, the interior relaxes to the stationary filter, and the pure-impact problems close in the two forms familiar from execution, block-plus-continuous and U-shaped.

Three restrictions bound the scope. The costs are quadratic, so nonlinear impact is outside the model. The explicit filters require a stationary Gaussian signal; for a general adapted signal, Theorem 1 gives the policy through the forecast curve but not as a transfer function. And the asset is one-dimensional; a portfolio version would call for a matrix Wiener–Hopf factorization, explicit only in special cases. On finite horizons the operator form (21) is complete, but sharp constants for the constraint boundary terms remain to be worked out.

A Proofs

Lemma 1. Causality of A_+ gives $(I-P_+)A_+P_+ = 0$, so A_+ and A_+^{-1} preserve the adapted subspace; taking adjoints with $A_- = A_+^*$ gives $P_+A_-(I-P_+) = 0$, so A_- and A_-^{-1} preserve the orthogonal complement. For adapted v, decompose $A_-v = P_+A_-v + (I-P_+)A_-v$; applying A_-^{-1}, the second image stays in the complement and P_+ annihilates it, so $P_+A_-^{-1}P_+A_-v = P_+v = v$. Hence for adapted u,

$$A_+^{-1}P_+A_-^{-1}\,(P_+AP_+)\,u = A_+^{-1}\big[P_+A_-^{-1}P_+A_-\big]A_+u = A_+^{-1}A_+u = u,$$

using $P_+A_+u = A_+u$; the reverse composition is the identity by positivity of P_+AP_+ on the adapted subspace. For the unbounded power-law factors the identity holds on the dense domain of signals with $\int(1+\omega^2)S_\alpha(\omega)/\hat{q}(\omega)\,d\omega < \infty$; any $\eta > 0$ or $\lambda > 0$ gives bounded inverses. $\square$

Theorem 1. Strict convexity ($\hat{n} > 0$ almost everywhere) makes (4) necessary and sufficient with a unique solution, which Lemma 1 with $A = N$ inverts. The forecast-curve form follows from conditional Fubini: N_-^{-1} has a deterministic anticausal kernel, so $\mathbb{E}_s[(N_-^{-1}\mu)(s)]$ integrates that kernel against $\mathbb{E}_s[\mu_r] = \bar{\mu}(s,r)$ for $r \geq s$. The rate form (17) follows by differentiating: $\partial_t \circ N_+^{-1} = Q_+^{-1}$, and on forecast curves $Q_-^{-1} = N_-^{-1}\circ(-\partial_s)$ with $-\partial_s\bar{\alpha}(s,\cdot) = \bar{\mu}(s,\cdot)$, so the whitened forecasts coincide. Applying N to the policy gives $Nx^\star = \mu - \xi^\star$ with $\xi^\star = N_-(I-P_+)N_-^{-1}\mu$; since N_- is anticausal, $\xi^\star$ is strictly anticausal, hence $P_+\xi^\star = 0$ and $\mathbb{E}\langle x, \xi^\star\rangle = 0$ for adapted x, giving (10). $\square$

Theorem 2. The whitened signal $N_-^{-1}\mu$ is the stationary filter of $\dot{W}$ with symbol $h = \hat{\psi}\,\hat{n}_-^{-1}$. Conditioning at s annihilates innovations after s, so P_+ truncates the moving-average kernel at lag zero and acts as $[\cdot]_+$; composing with N_+^{-1} gives (11). For $\lambda > 0$, $\hat{n}$ is bounded below, so h and $\hat{\pi}$ lie in L^2; the decay hypothesis $\int(1+\omega^2)S_\mu/\hat{n}\,d\omega < \infty$ places h in a Sobolev class on which the half-line indicator multiplies boundedly (Lions and Magenes 1972, Ch. 1), giving $\hat{\chi} \in L^2$. The first-order condition and adaptedness give $v = \frac{1}{2}\mathbb{E}[x_t^\star \mu_t]$; by the Itô isometry with $\overline{\hat{\psi}} = \hat{n}_+\overline{h}$, $\mathbb{E}[x^\star\mu] = \frac{1}{2\pi}\|\,[h]_+\,\|^2$, which is (12). Replacing $[\cdot]_+$ by the identity gives $v_{\mathrm{ant}} = \frac{1}{4\pi}\int S_\mu/\hat{n}$, and orthogonality of causal and anticausal parts gives $v_{\mathrm{ant}} - v = \frac{1}{4\pi}\|\,[h]_-\,\|^2 = \frac{1}{2}\mathbb{E}\langle\xi^\star, N^{-1}\xi^\star\rangle$. $\square$

Ornstein–Uhlenbeck formulas. With $\hat{\psi} = \sigma\theta/(\theta - i\omega)$, the combination $h - \hat{\psi}/\Phi(\theta) = \sigma\theta[\hat{n}_-^{-1}(\omega) - \hat{n}_-^{-1}(-i\theta)]/(\theta - i\omega)$ is anticausal: the pole at $\omega = -i\theta$ cancels, $\hat{n}_-^{-1}$ is analytic in the lower half-plane, and $\hat{n}_-(-i\theta) = \hat{n}_+(i\theta) = \Phi(\theta)$ by conjugate symmetry. Hence $[h]_+ = \hat{\psi}/\Phi$, giving (13), and with $\|\hat{\psi}\|^2 = \pi\sigma^2\theta$, $v = \sigma^2\theta/4\Phi^2$. For a Markov signal of order d, $[h]_+$ is the sum of the d residues, giving the differential operator $P(\partial)$. $\square$

Power-law formulas. From $\hat{g}(\omega) = c_\beta|\omega|^{\beta-1}$ the rate symbol $\hat{q} = \gamma c_\beta|\omega|^{\beta-1}$ has outer factor $(\gamma c_\beta)^{1/2}(\mp i\omega)^{-\nu}$, $\nu = (1-\beta)/2$, the symbols of $(\gamma c_\beta)^{1/2}I_\pm^\nu$, with inverses $D_\pm^\nu$ of symbol $(\mp i\omega)^\nu$; substituting in (17) gives (18). For the position, $\hat{n}_+ = (\gamma c_\beta)^{1/2}(-i\omega)^{(1+\beta)/2}$ in (13) gives (19) and spectral density $|\omega|^{-(1+\beta)}S_\mu$ up to a constant, integrable at the origin iff $S_\mu = o(|\omega|^\beta)$. For the value, the rate forms give $v = \frac{1}{4\pi}\|\,[h_\alpha]_+\,\|^2$ and $v_{\mathrm{ant}} = \frac{1}{4\pi}\|h_\alpha\|^2$ with $h_\alpha = (\gamma c_\beta)^{-1/2}(-i\omega)^\nu\hat{\varphi}$, whose plus-part and full norms differ by the factor $\sin(\pi\beta/2)$ from the argument of $(-i\omega)^{\beta-1}$ across the two half-lines; with $\hat{\varphi} = \sigma/(\theta - i\omega)$ this gives $v = \sigma^2\theta^{-\beta}/4\gamma c_\beta$ and (20). $\square$

Proposition 1. The finite-horizon policy (21) differs from the whole-line one through the terminal-anchored weight and the truncation of the factor kernels to $[0,T]$. For the power-law factor (24) the truncation cuts the Marchaud tail at distance $d(t)$, contributing $d(t)^{-\nu}$; the weight deviation $((T-s)/(T-t))^\nu = 1 + O(d(t)^{-1})$ is subdominant. For a rational symbol the causal factor $\hat{n}_+^{-1}$ has kernel decaying as $e^{-b_1\tau}$ at its slowest zero b_1, so truncation at distance $d(t)$ costs $e^{-b_1 d(t)}$; equivalently the finite-horizon feedback gains relax to their stationary values at rate $2b_1$. $\square$

Volterra factors on $[0,T]$. For the exponential kernel, $(\kappa^2 - \partial_t^2)(E_\kappa f) = 2\kappa f$ on $(0,T)$ with $(E_\kappa f)'(0) = \kappa(E_\kappa f)(0)$ and $(E_\kappa f)'(T) = -\kappa(E_\kappa f)(T)$; since $\kappa^2 - \partial_t^2 = (\kappa + \partial_t)^*(\kappa + \partial_t)$, the inverse factors are (23), with the two Robin conditions realized as rank-one endpoint atoms and

C_+ of Green's function $\sqrt{2\gamma\kappa}\, e^{-\kappa(t-s)}$; this was checked numerically against the reverse-Cholesky factor of $e^{-\kappa|t-s|}$. For the power-law kernel, direct integration of (24) gives $\int_0^T c_+(u,t)c_+(u,s)\,du = \gamma c_\beta |t-s|^{-\beta}$ through a Beta-type integral, using $2\nu = 1-\beta$ and $c_\beta = 2\Gamma(1-\beta)\sin(\pi\beta/2)$, so $C_- C_+ = G_T$; the placement of C_+ on the right of (21) forces the terminal anchor. $\square$

Acknowledgements. The author thanks CFM colleagues, and in particular Jean-Yves Audibert, Antoine Bambade, Raphael Benichou, Jean-Philippe Bouchaud, Raphael Maillet, Iacopo Mastromatteo and Marc Potters, for useful discussions. Research and write-up were assisted by large language models. Verification scripts and figure code are available from the author.

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